D.19 Hessian

Here, we compute the Hessian for several functions.

Lemma D.45 (Hessian of $u(x)$, Lemma 5.9 in [105]). *Provided that the following conditions are true*

- *Given a matrix $A \in \mathbb{R}^{n \times d}$.*
- *For every $i \in [d]$, let $A_{*,i} \in \mathbb{R}^n$ denote the i-th column of matrix A.*
- *Let $\circ$ denote hadamard product.*

Then, we have, for each $i \in [d]$

- *Part 1.*
$$\frac{\mathrm{d}^2 \exp(Ax)}{\mathrm{d}x_i^2} = A_{*,i} \circ u(x) \circ A_{*,i}$$
- *Part 2.*
$$\frac{\mathrm{d}^2 \exp(Ax)}{\mathrm{d}x_i \mathrm{d}x_j} = A_{*,j} \circ u(x) \circ A_{*,i}$$

Lemma D.46 (Lemma 5.10 in [105]). *Provided that the following conditions are true*

- *We define $\alpha(x)$ as Definition D.36.*
- *For every $i \in [d]$, let $A_{*,i} \in \mathbb{R}^n$ denote the i-th column of matrix A.*
- *Let $\circ$ denote hadamard product.*

Then, we have

- *Part 1.*
$$\frac{\mathrm{d}^2 \alpha(x)}{\mathrm{d}x_i^2} = \langle u(x), A_{*,i} \circ A_{*,i} \rangle$$
- *Part 2.*
$$\frac{\mathrm{d}^2 \alpha(x)}{\mathrm{d}x_i \mathrm{d}x_j} = \langle u(x), A_{*,i} \circ A_{*,j} \rangle$$
- *Part 3.*
$$\frac{\mathrm{d}^2 \alpha(x)}{\mathrm{d}x^2} = A^\top \operatorname{diag}(u(x)) A$$

D.20 Hessian is Positive Definite

It is well known that in literature [104, 105, 107], the Hessian H of loss function can be written as $A^\top (B(x) + W^2) A$ for some matrix function $B(x) \in \mathbb{R}^{n \times n}$ (for example see explanation in Section 5.10 in [105]). In this section, we show that Hessian is positive definite.

Lemma D.47. *If the following conditions hold*

- *Given matrix $A \in \mathbb{R}^{n \times d}$.*
- *We define $L_{\mathrm{sparse}}(x)$ as Definition D.41.*
- *We define $L_{\mathrm{sparse}}(x)$ Definition D.40.*
- *We define $L_{\mathrm{sparse}}(x)$ Definition D.39.*
- *Let $L(x) = L_{\exp}(x) + L_{\mathrm{sparse}}(x) + L_{\mathrm{reg}}(x)$.*

- *Let $A^\top (B(x) + W^2)A$ be the Hessian of $L(x)$*
- *Let $W = \mathrm{diag}(w) \in \mathbb{R}^{n \times n}$. Let $W^2 \in \mathbb{R}^{n \times n}$ denote the matrix that i-th diagonal entry is $w_{i,i}^2$.*
- *Let $\sigma_{\min}(A)$ denote the minimum singular value of A.*
- *Let $l > 0$ denote a scalar.*

Then, we have

- *Part 1. If all $i \in [n]$, $w_i^2 \geq 20 + l/\sigma_{\min}(A)^2$, then*
$$\frac{\mathrm{d}^2 L}{\mathrm{d}x^2} \succeq l \cdot I_d$$
- *Part 2 If all $i \in [n]$, $w_i^2 \geq 200 \cdot \exp(R^2) + l/\sigma_{\min}(A)^2$, then*
$$(1 - 1/10) \cdot (B(x) + W^2) \preceq W^2 \preceq (1 + 1/10) \cdot (B(x) + W^2)$$

Proof. The entire proof framework follows from [104, 105, 107], in the next few paragraphs, we mainly explain the difference.

The $B(x)$ based on L_{sparse} is $\mathrm{diag}(u(x))$. Note that it is obvious that
$$\mathrm{diag}(u(x)) \succeq 0.$$
From the upper bound size, we know that
$$\begin{aligned} \mathrm{diag}(u(x)) &\preceq \|u(x)\|_\infty \cdot I_n \\ &\preceq \exp(R^2) \end{aligned}$$
where the last step follows from Proof of Part 0 in Lemma 7.2 in [105].

To prove Part 1, following from [104, 105], we only use the lower bound of $\mathrm{diag}(u(x))$. By putting things together, we get our results.

To prove Part 2, we follow from [107] and use both the upper bound and lower bound of $\mathrm{diag}(u(x))$.

□

D.21 Hessian is Lipschitz

In this section, we show Hessian is Lipschitz.

Lemma D.48. *If the following conditions hold*

- *Let $H(x) = A^\top (B(x) + W^2)A$ denote the Hessian of L.*

Then, we have

- $\|H(x) - H(y)\| \leq n^2 \exp(40R^2)\|x - y\|_2$

Proof. The entire proof framework follows from [104, 105, 107], in the next few paragraphs, we mainly explain the difference.

Note that the $B(x)$ based on $L_{\exp} + L_{\mathrm{reg}}$ have been proved by [105]

We only need to prove $B(x)$ based on L_{sparse} and add them together.

Note that $B(x)$ based on L_{sparse} is in fact $\mathrm{diag}(u(x))$.

Using Lemma 7.2 in [105], we know that
$$\begin{aligned} \|\mathrm{diag}(u(x)) - \mathrm{diag}(u(y))\| &\leq \|u(x) - u(y)\|_2 \\ &\leq 2\sqrt{n} R \exp(R^2)\|x - y\|_2 \end{aligned}$$
where the last step follows from Part 1 in Lemma 7.2 in [105].

Thus, putting things together, we complete the proof. □

D.22 Greedy Type Algorithm

In this section, we propose a greedy-type algorithm (based on the approximate Newton method) to solve the optimization problem.

Algorithm 4 A greedy type algorithm.

procedure $\textsc{OurIterativeMethod}(A \in \mathbb{R}^{n \times d}, b \in \mathbb{R}^n, w \in \mathbb{R}^n, \epsilon, \delta)$
  Initialize x_0
  $T \leftarrow \log(\|x_0 - x^*\|_2/\epsilon)$ ▷ Let T denote the number of iterations.
  for $t = 0 \to T$ **do**
    $D \leftarrow B_{\mathrm{diag}}(x_t) + \mathrm{diag}(w \circ w)$
    $\widetilde{D} \leftarrow \textsc{SubSample}(D, A, \epsilon_1 = \Theta(1), \delta_1 = \delta/T)$
    Compute gradient g exactly
    Get the approximate Hessian $\widetilde{H}$ by computing $A^\top \widetilde{D} A$
    Update x_{t+1} by using the Newton step $x_t + \widetilde{H}^{-1} g$
  end for
  $\widetilde{x} \leftarrow x_{T+1}$
  return $\widetilde{x}$
end procedure

Theorem D.49. *Given matrix $A \in \mathbb{R}^{n \times d}$, $b \in \mathbb{R}^n$, and $w \in \mathbb{R}^n$.*

- *We use x^* to denote the optimal solution of*

$$\min_{x \in \mathbb{R}^d} L_{\exp} + L_{\mathrm{sparse}} + L_{\mathrm{reg}}$$

  that

  - $g(x^*) = \mathbf{0}_d$*, where g denotes the gradient function.*
  - $\|x^*\|_2 \leq R$.

- *Suppose that $R \geq 10$, $M = \exp(\Theta(R^2 + \log n))$, and $l > 0$.*

- *Assume that $\|A\| \leq R$. Here $\|A\|$ denotes the spectral norm of matrix A.*

- *Suppose that $b \geq \mathbf{0}_n$ and $\|b\|_1 \leq 1$. Here $\mathbf{0}_n$ denotes a length-n vector where all the entries are zeros. (Here $b \geq \mathbf{0}_n$ denotes $b_i \geq 0$ for all $i \in [n]$)*

- *Assume that $w_i^2 \geq 200 \cdot \exp(R^2) + l/\sigma_{\min}(A)^2$ for all $i \in [n]$. Here $\sigma_{\min}(A)$ denotes the smallest singular value of matrix A.*

- *Let x_0 denote an starting/initial point such that $M\|x_0 - x^*\|_2 \leq 0.1l$.*

- *We use to $\epsilon \in (0, 0.1)$ represent our accuracy parameter.*

- *We use $\delta \in (0, 0.1)$ to represent failure probability.*

There is a randomized algorithm that

- *runs $\log(\|x_0 - x^*\|_2/\epsilon)$ iterations*

- *spend*

$$O((\mathrm{nnz}(A) + d^\omega) \cdot \mathrm{poly}(\log(n/\delta))$$

  time per iteration,

- *and finally outputs a vector $\widetilde{x} \in \mathbb{R}^d$ such that*

$$\Pr[\|\widetilde{x} - x^*\|_2 \leq \epsilon] \geq 1 - \delta.$$